

A Visual and Spatial Analysis of Pricing Dynamics in London's Airbnb Market

1. Aim

London's Airbnb market plays a significant role in the city's housing and tourism economy; however, it is often considered a homogenous system rather than a highly fragmented market influenced by the type of host and the spatial variation in pricing. Research suggests that a small number of multi-listing hosts are responsible for a large share of the total supply in Airbnb. As demonstrated by A. Stewart Fotheringham, Brunsdon, and Charlton (2002), factors like location and reputation have different effects on property pricing within Central London compared to Outer London (insideairbnb.com, n.d.). Inspecting these variations is vital to avoid the true economic dynamics of a housing market being misinterpreted, resulting in inaccurate valuation models and the development of ineffective housing policies.

This report investigates the structural segmentation and spatial dynamics in London's Airbnb market. It addresses three main questions:

1. Do commercial operators form a distinct market segment?
2. Do Airbnb price determinants vary across the city?
3. Does a spatially differentiated model outperform the standard global model in prediction strategy?

Visual analytics methods like K-means clustering and Geographically Weighted Regression (GWR) were used to uncover hidden structural distortions and local pricing trends in London's Airbnb market.

2. Visual Analytics

2.1 Workflow Diagram

Figure 1 shows the five phases of analysis. Phase 1 consists of data cleaning and various exploratory data analysis (EDA) methods. Phase 2 uses t-SNE and k-means clustering to perform structural separation of commercial operators from the private hosts. Phase 3 tests global OLS regression and residual mapping to evaluate spatial non-stationarity. Phase 4 applies a Geographically Weighted Regression (GWR) to partition the city into different pricing regimes. Phase 5 applies k-means clustering on the coefficients and the spatial model for market segmentation.

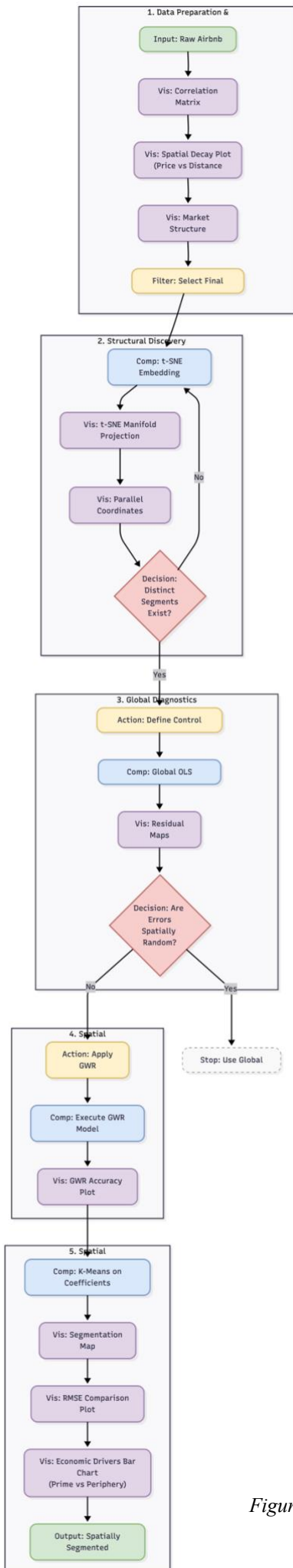


Figure 1: Workflow Diagram

2.2 Data Preparation & EDA

The raw London Airbnb dataset was cleaned rigorously; listings with missing values and duplicates were dropped, and highly skewed variables like *price* were log-transformed to ensure a normal distribution. A key spatial variable, *Distance_to_Center*, was calculated using Charing Cross, which is known as the geographic centre of London, to perform the spatial gradient analysis.

The descriptive statistics table (Figure 2) gives a numerical representation of the skewness, with a high variation from the median *price* of £131 to the mean *price* of £192.82. The high standard deviation explains the choice to model with *log-price*. The standard deviation also indicates the possible presence of large-scale operators, with a *calculated_host_listings_count* standard deviation of 40.80.

Figure 2: Descriptive Statistics

	Mean	Median	Std
id	N/A	N/A	N/A
latitude	N/A	N/A	N/A
longitude	N/A	N/A	N/A
accommodates	3.53	3	2.17
bathrooms	1.34	1	0.66
bedrooms	1.53	1	0.96
beds	1.96	2	1.38
price	192.82	131	862.52
minimum_nights	4.18	2	13.70
availability_365	203.24	221	120.27
number_of_reviews	34.79	13	64.15
review_scores_rating	4.67	4.82	0.51
calculated_host_listings_count	16.44	3	40.80
reviews_per_month	1.28	0.83	1.42
Distance_to_Center	7.39	6.01	4.90
Log_Price	4.88	4.88	0.75

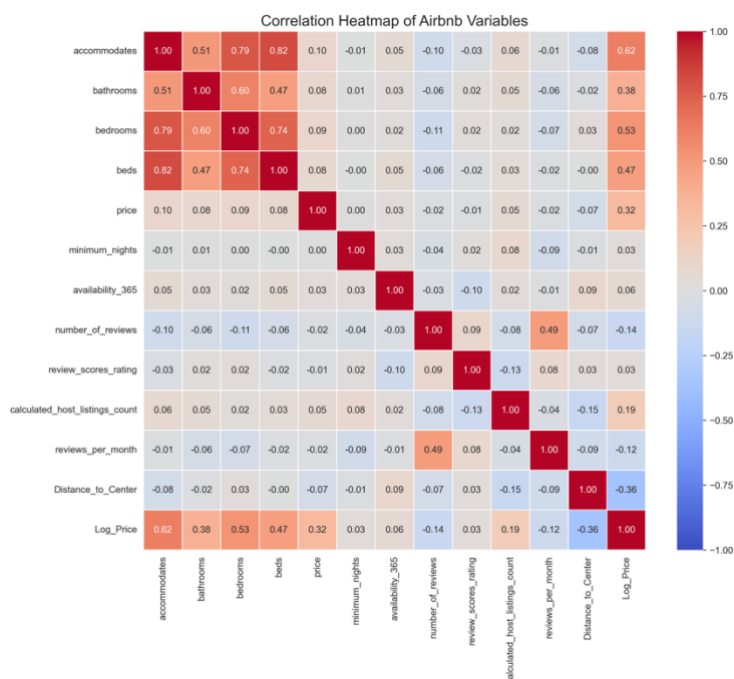
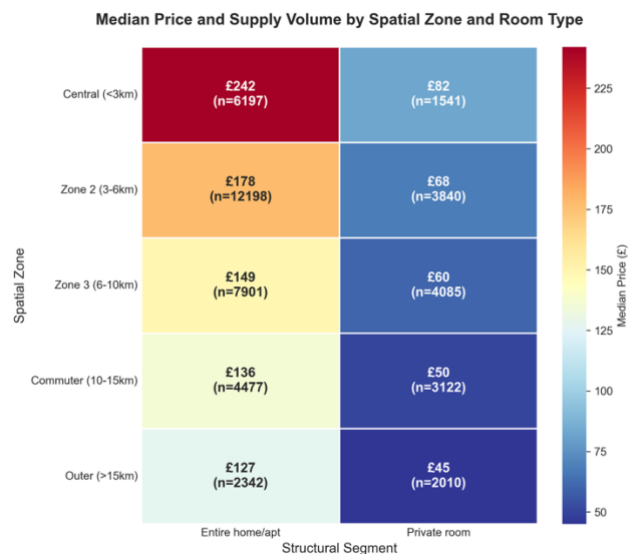
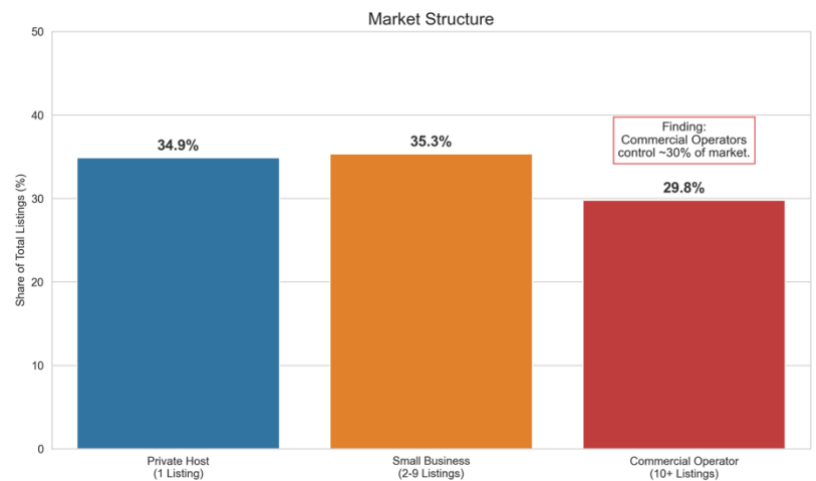


Figure 3 shows a correlation matrix for identifying the large multicollinearity among various structural characteristics; to mitigate this, the *beds* variable was dropped since it was highly correlated to *bedrooms* (Pearson $r = 0.74$).

Figure 3: Correlation Heatmap

In Figure 4, the Market Structure Bar Chart shows that the 'commercial' operators (those who have more than 10 properties available) dominate the market by controlling 29.8% of the total available units, challenging the idea of a 'sharing economy'.

Figure 4: Bar Chart – Airbnb Market Structure



The heatmap in Figure 5 illustrates a significant decrease in prices as you move from Central London (£242) to Outer Zones (£127) with entire apartments, while private rooms only decline marginally from £82 to £45. Therefore, it indicates that location affects the value of higher-priced properties significantly compared to lower-priced offerings.

Figure 5: Heatmap - Median price and supply volume by spatial zone and room type

The histogram in Figure 6 shows an extensive ceiling effect (mean = 4.67), where most of the listings are concentrated above the 4.5-star mark, thereby limiting the ability of this variable to discriminate between listings. A distinct spike at 5 stars indicated positivity bias, which implies that a perfect rating is the market's general standard. The homogeneity of the scores suggests that the summary score provides little value in predicting price.

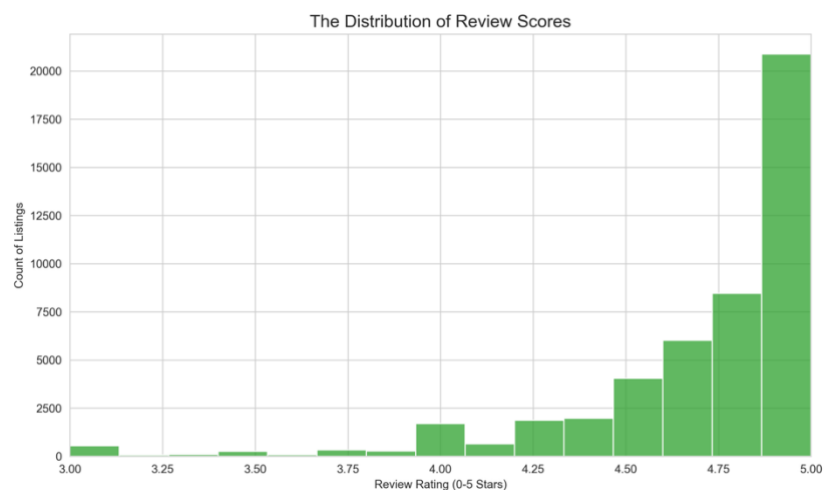
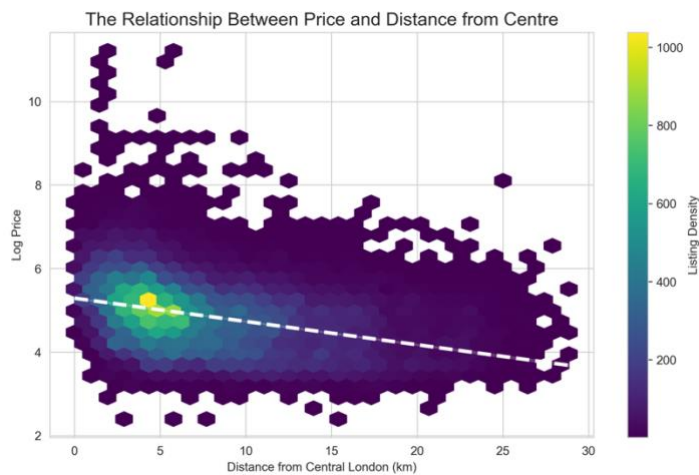


Figure 6: Histogram – distribution of review_scores_rating



The hexbin plot in Figure 7 shows a significant concentration of yellow-green hexagons in the lower left, indicating that most affordable options cluster centrally. However, as you move away from the centre, there is a considerable decrease in price variance as the distance from the centre increases (spatial heteroscedasticity).

Figure 7: Hexbin plot – relationship between price and distance from centre.

2.3 Structural Discovery: Market Segmentation & Clustering

The t-SNE method was used to analyse the relationship between price, availability and the number of listings for host types. While this method was strictly exploratory, it is a random method and may produce inaccurate representations of true distances.

However, the manifold projection in Figure 8 clearly shows a definable pattern to the data, which does not appear to have occurred randomly. Rather than existing as a homogeneous cloud, these data points represent unique islands. Visually, this indicates that there are multiple segments (or sub-groups) of hosts operating under very different rules.

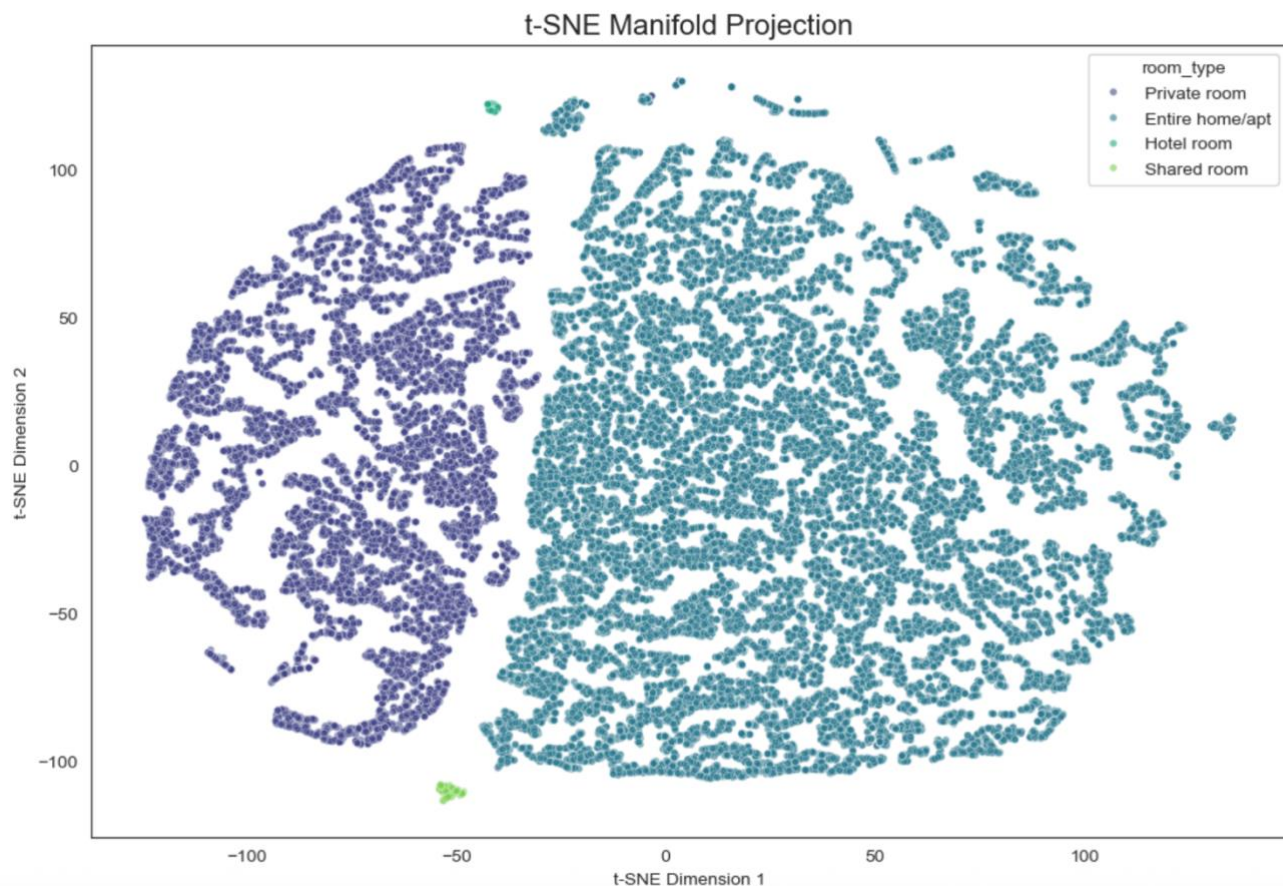


Figure 8: t-SNE Manifold Projection

To test and mathematically group the data to support the hypothesis, K-means clustering was implemented. The parallel coordinates plot (Figure 9) illustrates the characteristics of these groups and how the data in each cluster is arranged. The plot clearly illustrates differences between host behaviour:

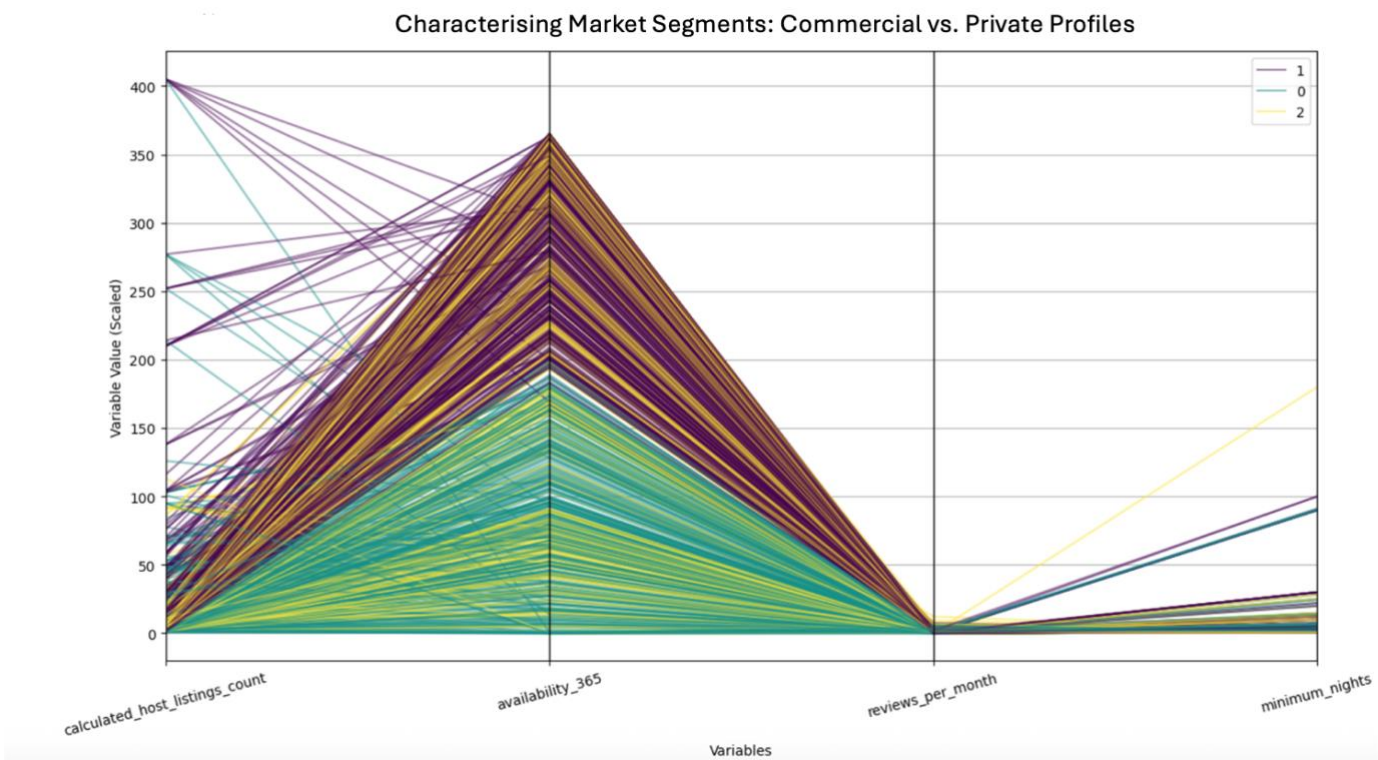


Figure 9: Parallel Coordinates plot

Cluster 1 represents the Commercial Operators, characterised by a high number of listings and are available all year long. These hosts typically operate through distributed hotels rather than sharing their homes. Cluster 0 represents Private Hosts; they have a single listing with inconsistent availability, which is indicative of a typical non-commercial homeowner. Finally, Cluster 2 represents Long-Term Lets, experiencing a notable increase in *minimum_nights*, likely a result of hosts transitioning to long-term rentals to circumvent restrictions imposed by short-term rental regulations.

This information indicates that London has multiple markets, as opposed to a single market, and is characterised by a 'dual-system' economy. Treating these distinct groups as a homogenous group would result in significant bias. By identifying two different business perspectives (operationally) within Phase 3, the pricing models will control these different segments explicitly.

2.4 Global Modelling & Diagnostics

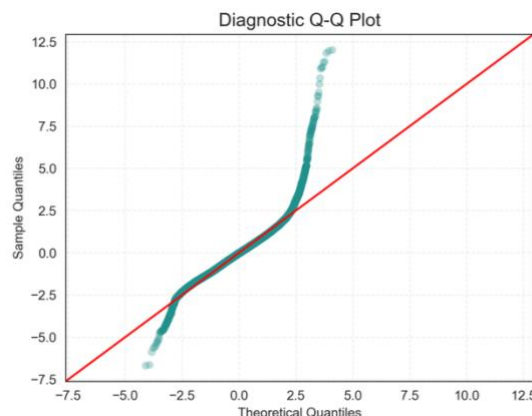
The baseline for the factors that drive the pricing of the property was developed by fitting an Ordinary Least Squares (OLS) Regression Model to the data. *Log_Price* was the dependent variable while the independent variables were *accommodates*, *bathrooms*, *bedrooms*, *review_scores_rating*, and *Distance_to_Center* to capture general spatial trends.

$$\text{Log_Price} = 4.1250 + 0.1519 \text{ accommodates} + 0.0510 \text{ bathrooms} + 0.1296 \text{ bedrooms} - 0.0510 \text{ Distance_to_Center} + 0.0706 \text{ review_scores_rating} + \epsilon$$

The Global Model achieved an adjusted R^2 value of 0.5, which indicates that 50% of the variance in *Log_Price* was explained by the structural variables. The coefficients indicate that *accommodates* and *bedrooms* are strong positive influences, while the distance to the centre has a significant negative coefficient (-0.051). Therefore, with every 1 unit increase in distance from Charing Cross, the price decreases by approximately 5%.

However, visual diagnostic evaluation identifies significant weaknesses in the global model. Large deviations from the diagonal line on the tails of the Q-Q plot in Figure 10 imply that the model's residuals are not normally distributed. In practice, this indicates that the global model does not effectively value the extremes of the market and consistently misjudges luxury and budget accommodations.

Figure 10: OLS Regression Q-Q Plot



Distribution of Price Residuals by Distance to Centre

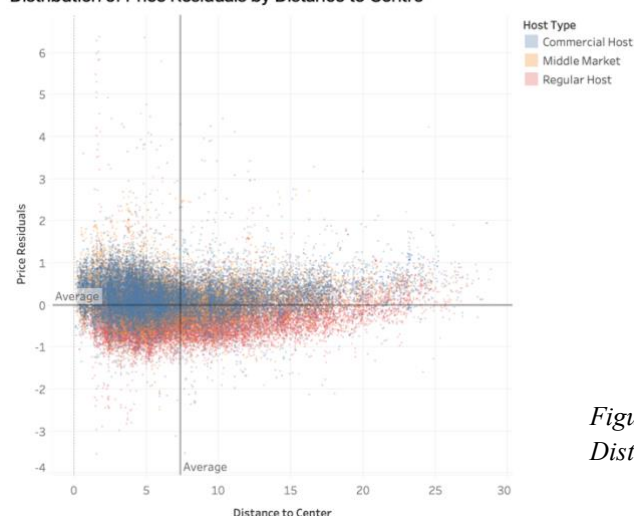


Figure 11 indicates serious heteroscedasticity. The distinct funnel shape of the scatter plot and the dispersion of residuals at short distances (0 to 5km from Charing Cross). This indicates that the model performs poorly by making large over- and underpredictions regarding property prices in Central London. Conversely, as distance increases, the cloud narrows, suggesting that the model becomes increasingly accurate in predicting property values in the suburbs. In summary, the variance in property prices across different areas of London is very volatile, with Central London being quite difficult to forecast in comparison to suburban areas.

Figure 11: Scatter plot – Distribution of Price Residuals by Distance_to_Centre

Moran's I statistic value of 0.2754 ($p < 0.01$) confirmed a significant positive spatial autocorrelation of price residuals, indicating that instead of being randomly distributed, model errors are distributed geographically. The large blue area in Figure 12 represents a central core in London, where prime assets have been systematically under-predicted, while most surrounding outer areas show clusters of negative residuals (where prices have been over-predicted, shown in red). This confirms that price determinants vary spatially and therefore cannot be captured adequately by a single global regression point estimate on its own.

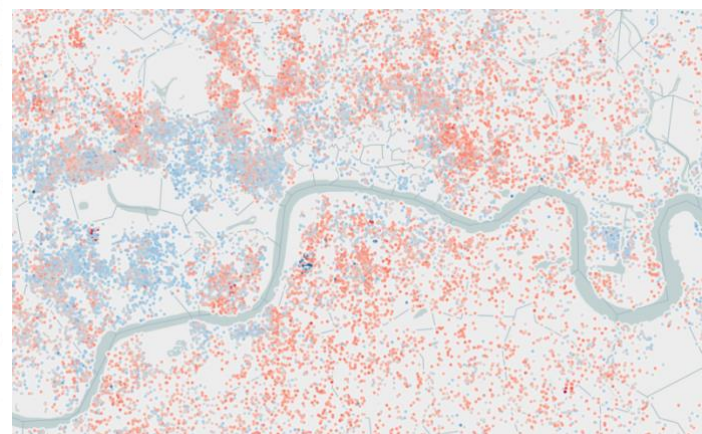
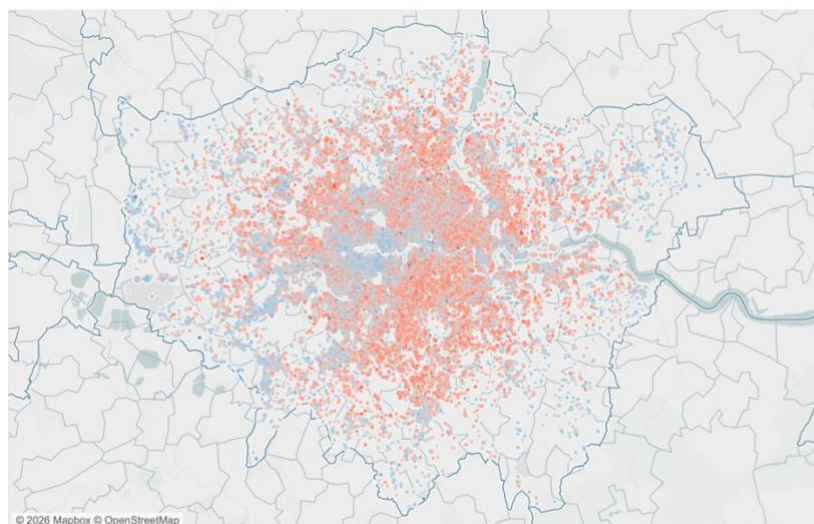


Figure 12: Spatial Distribution of Price Residuals in London (left) and Central London (right)

2.5 Geographic Weighted Regression (GWR)

A Geographically Weighted Regression (GWR) was executed to produce local regression equations at each location to model the spatially variable price determinants identified in the diagnostics. Converting to local modelling improved the overall fit of the model by a large margin, as the Adjusted R^2 increased from 0.50 (OLS) to 0.640 (GWR), accounting for an additional 14% of *Log_Price* variance.

The GWR accuracy plot (Figure 13) visually confirms this, showing the predicted GWR values closely align with the perfect fit line, indicating that the local model provides far superior modelling of market variance compared to the global baseline.

Figure 13: GWR accuracy plot

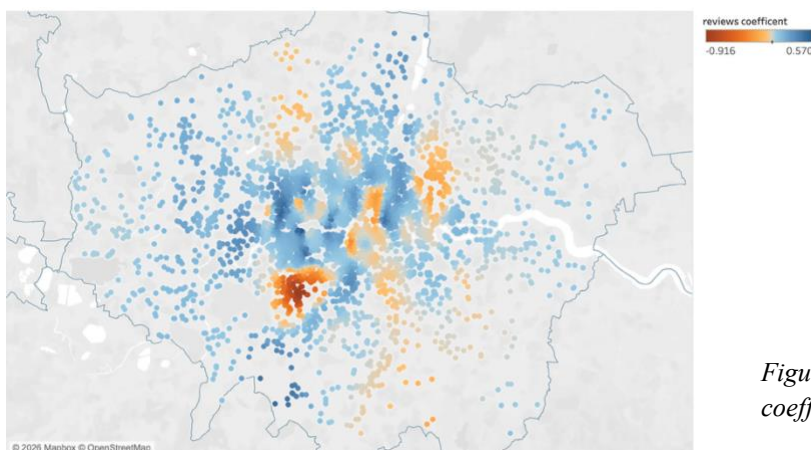


Figure 14 indicates how the GWR approach reveals a duality of reputation. Specifically, whilst review scores in core areas are positively related to price (blue) and hence a sign of trust for high-risk tourism locations, periphery areas have the opposite relationship (orange), with high review counts indicating high turnover utility rather than quality for budget-sensitive markets.

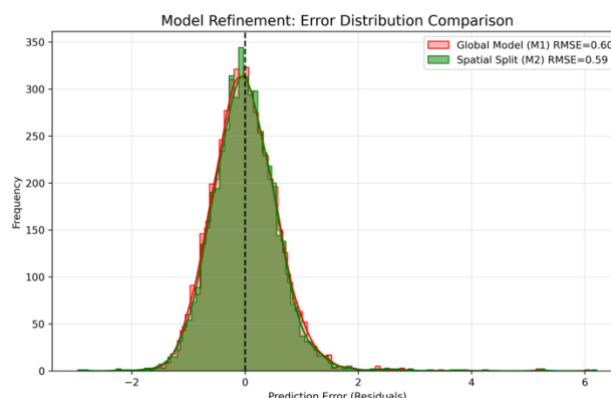
Figure 14: Spatial distribution of *review_scores_rating* coefficients (GWR)

2.6 Spatial Segmentation

K-means clustering was executed once again based on Review Coefficients, Latitude and Longitude to create a spatial model. Kensington was used as a reference anchor point to find the "Prime" cluster, effectively grouping areas based on economic behaviour as well as geographical location.

Figure 15 demonstrates how the spatial model (green) has a tighter error spread than the global baseline (red). Therefore, segmentation decreased the Global Root Mean Square Error (RMSE) from 0.60 to 0.59, which shows that using one city-wide equation (the global model) could not accurately identify all the characteristics of the property market in London.

Figure 15: Error Distribution Comparison, Global model (red) vs Spatial model (green)



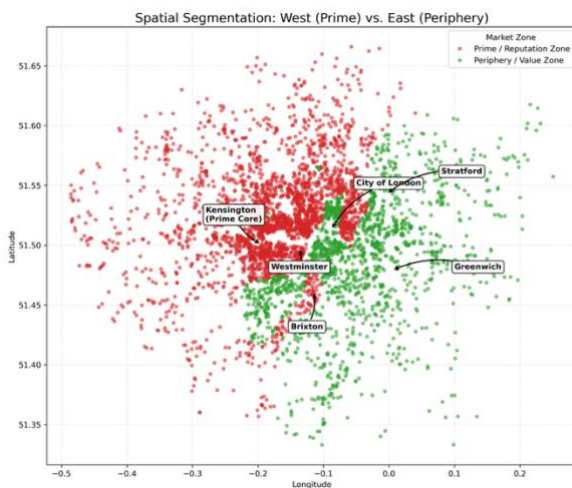


Figure 16 shows the “Prime” (Red) and “Periphery” (Green) market segments explicitly separated. Both markets have clear cuts due to their clustering around spatially-smoothed GWR coefficients instead of their raw price volatility. The geographic split demonstrates a clearly defined point where the basis for market pricing changes from a review-based pricing in the West to utility-driven pricing in the East.

Figure 16: Scatter plot map - Spatial Segmentation

The Coefficient Comparison (Figure 17) provides further evidence that drivers of economic activity differ on either side of this divide. The negative coefficient in *review_scores_rating* shows that it serves as a much greater positive (+0.103) driving force in the Prime section than it does in the Peripheral (-0.031) section. Another important distinction is that Prime areas are willing to price properties higher than Peripheral areas for more *bedrooms* (+0.063 versus +0.037).

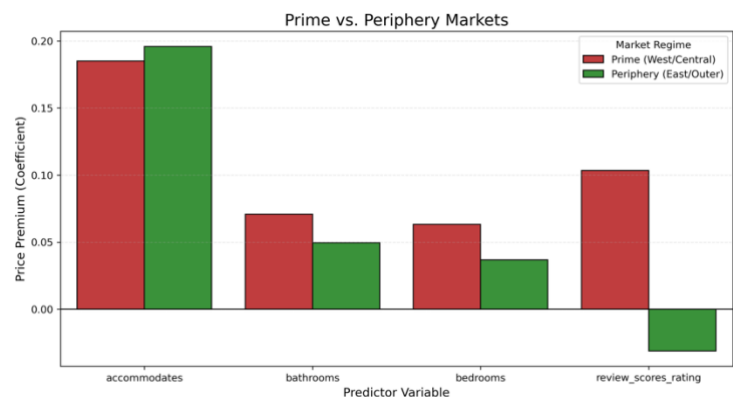


Figure 17: Bar Chart comparing Prime vs Periphery

3. Findings

Regarding market segmentation, the initial global residual mapping revealed where errors (spatial bias) occurred; therefore, the results demonstrate that commercial operators operate in a substantially different environment than those within the Prime industry sector. GWR confirmed that there are two primary segments of the market: operators that depend on their review for earnings (Prime segment) and those who benefit from a significant volume of sales (utility-driven segment).

Focusing on the spatial variability of price determinants, it is concluded that geographic location determines the price determinant’s value, as shown by Figure 17. Reputation (*review_scores_rating*) holds different values spatially; in the West, better reviews cause higher property pricing. In the Periphery regions, price signals based on reviews lose their correlation with price, confirming that pricing decisions made in the outer zones are determined by utility logic, rather than reviews.

Regarding prediction models, the spatial model was superior to the global baseline. The GWR increased the explanatory power of the model, while breaking the city into two zones, prime and periphery, also resulted in lower global RMSE. Therefore, it is concluded that a spatial model can produce better valuation estimates compared to a single model that applies to an entire city.

4. Critical Reflection

4.1 Evaluation

The study successfully investigated spatial variations in London's property market by using visual analytics not just to present findings but as a way to generate hypotheses. Initially, the single global regression of property prices on the independent variables created a misleading average coefficient, which obscured the actual economic factors that affect property prices in London. The spatial visualisation of GWR residuals and coefficient surfaces was the key method in detecting non-stationarity of price determinants. Specifically, Figure 17 provided the most solid empirical justification for rejecting the homogenous market hypothesis. Without having the capacity for interactive mapping and segmentation of these local coefficients, the structural Prime and Periphery divide would have been statistically invisible.

4.2 Methodological Critique

The distinct segmentation in 2.5 was produced through bandwidth selection in the GWR. This parameter averages the neighbouring observations, leading to smooth local trends and a perception of a continuous Prime-Periphery border. Figure 16 concludes a significant difference between the Prime and Periphery GWRs; however, the market structure in reality is more fragmented than the visualised data would imply.

The clustering process in 2.6 relied on domain knowledge (using Kensington as a reference point) instead of total, unsupervised learning, meaning that the workflow cannot be implemented in other cities where such qualitative variables do not exist. Finally, by using cross-sectional data, the findings are restricted to that time frame, preventing detection of whether those boundaries are changing over time due to gentrification.

4.3 Wider Implications

This research challenges how rental accommodation platforms are set up, specifically with regard to how algorithms rank properties that receive high reviews. While most views on ranking properties are based on a universal logic of the higher the reviews, the greater the value, this research demonstrates that the demand for reviews is conditioned by spatial location. In utility-driven segments, price sensitivity negates the ranking benefit derived from reviews. While these platforms may be efficiently optimising for tourists, but inefficient for the utility-driven consumers.

Using this spatial-economic clustering framework offers a strong alternative to the "black box" machine learning methods currently in use. By showing that a segmented linear model is a better predictor than a simple global model, this research advocates for an overall strategy of reduced complexity by identifying each of the spatial regimes before proceeding with model training. This will help balance predictive accuracy with interpretability.

4.4 Future Work

Future research can build upon this work by creating a foundation for spatial segmentation. The application of coefficient clustering in the GWR model outputs would therefore eliminate the necessity of using manual anchors such as Kensington, enabling transferability of the workflow into other urban environments (Kopczewska and Ćwiakowski, 2021). The inclusion of longitudinal data also enables tracking of the Prime-Periphery boundary throughout time, thereby creating a quantitative measure of gentrification. Finally, by providing a foundation for creating dynamic, adjustable platform architectures that rank algorithms based on local valuation regimes instead of applying a single global model.

5. Word Count

Section	Wordcount
1. Aim	199
2. Visual Analytics	1495
3. Findings	200
4. Critical Reflection	493

6. References

A. Stewart Fotheringham, Brunsdon, C. and Charlton, M. (2002). *Geographically Weighted Regression The Analysis of Spatially Varying Relationships* Wiley 2002. [online] Academia.edu. Available at: https://www.academia.edu/33626785/Geographically_Weighted_Regression_The_Analysis_of_Spatially_Varying_Relationships_Wiley_2002.

Inside Airbnb (2024). *Get the Data*. [online] insideairbnb.com. Available at: <https://insideairbnb.com/get-the-data/>.

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Kopczewska, K. and Ćwiakowski, P. (2021). Spatio-temporal stability of housing submarkets. Tracking spatial location of clusters of geographically weighted regression estimates of price determinants. *Land Use Policy*, 103, p.105292. doi:<https://doi.org/10.1016/j.landusepol.2021.105292>.

7. Appendix

In accordance with coursework guidelines, I declare that Google Gemini was utilised solely as a supplementary technical tool. I confirm that no assessment text was generated by AI, and all data interpretation, analysis of spatial regimes, and conclusions are my own original work. Usage was focused, iterative, and critically reviewed.

Summary of Application:

1. Used iteratively to troubleshoot specific syntax errors in geopandas and mgwr libraries.
2. Used to optimise code efficiency for the K-Means clustering loop.
3. Used to verify precise econometric definitions
4. Used to check my work to ensure it was hitting all the requirements in each section mark scheme, so that I could refine and check what was missing from my work.

Representative Prompts:

- “I am receiving a 'Singular Matrix' error when running GWR on the East London subset. Do not fix the code, but please explain the mathematical cause of this error in a local regression context so I can adjust my variables.”
- “I want to vectorise my Matplotlib plotting loop to improve rendering speed. Can you suggest a more efficient syntax structure?”
- “I am describing the loss of correlation between price and reviews. Is 'spatial decoupling' or 'non-stationarity' the more accurate econometric term for this specific phenomenon?”
- “Here is the mark Scheme for this section..... Does my write-up for this section cover all the requirements shown in the mark scheme?”
- “Explain how the GWR runs differently from the OLS regression”
- “How can I change the opacity and line thickness in the parallel coordinates to make the lines more distinct?”
- “I want to add subheadings in my sections. How can I effectively separate my sections into subheadings?”